Relationship of quantisation and token budgets for correctness in chain-of-thought language models.

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***Abstract*—Chain-of-Thought (CoT) large language models increase model capabilities for coverage of increasingly complex tasks by imitating thinking behavior, inspired by previous research on extending model capabilities through prompting techniques. In this report we outline our benchmarks for two models up to 1.5B parameter state-of-the-art reasoning models, aiming to identify the relationship between quantisation and thinking budgets for correctness in CoT-LLMs, which according to scaling laws scales as a power law with model size [1].**

Our results indicate that the accuracy of higher quantisation is indeed improved by allocating more compute to reasoning tokens of up to 512 at 8 and 4 bit quantisation levels, though comes at a significant request time tradeoff compared to adjusting for correctness via quantisation. Besides, we observe lower power costs per accuracy in reasoning compared to quantisation.

***Index Terms*—Large language models, Natural language processing, Quantization, Computational efficiency, Statistical analysis, Resource management.**

I. Introduction

The introduction of the attention mechanism [2] led to the development of the generative-pretrained transformer (GPT) model, iteratively improved in quality and efficiency through various techniques, which more recently includes reasoning behavior introduced through chain of thought prompting [3], [4] for complex tasks, and quantisation of the model weights for efficiency.

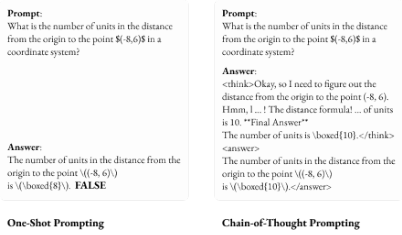

Fig. 1. Example for CoT on MATH dataset with Deepseek-R1-14B.

At the time of writing this paper, most of the LLMs topping the leaderboards [5]–[7] use therefore `<think></think>` tokens most commonly, which establish reasoning processes of various token lengths which imitate written thinking behavior for more complex reasoning, followed by an answer enclosed in `<answer></answer>` tokens. Between CoT models the respective usage may vary, and prompting may be imitated through using the bare words, but the most established method is separating the `<think>`/`<answer>` tokens from the regular tokens extracted from text-form training material by the tokenizer.

In Deepseek-R1 [8] such behavior is introduced through labeled reasoning as supervised finetuning (SFT), reinforcement learning (RL) without or with human feedback (RLHF) [9].

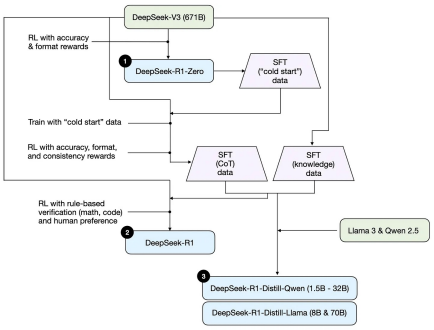

Fig. 2. Development process of the three Deepseek-R1 reasoning models [9].

Due to existing hardware constraints for offline-models, quantisation is an attractive option to deploy small- to mid-sized models on edge devices by reducing the bits of the model weights. Various quantisation methods emerged, ie. GPTQ which accounts for error on a calibration set to protect, to a certain degree, their capabilities [10].

Nevertheless, accuracy still degrades, which raises the question whether the benefit from low RAM- and storage-requirements of strongly quantised LLMs, is outweighed by the gains in accuracy through allocating larger token-budgets to restore performance in time-flexible scenarios.

We use GPTQModel [10], VLLM [11] and one NVIDIA V100 GPU of the Volta series to identify the relationship between quantisation and thinking token budgets, aiming to find the maximum improvement margin, which can be achieved on complex tasks.

We expect the generation time to significantly increase linearly with the token budgets, along with accuracy according to scaling laws until the information stored in the weights is insufficient to sufficiently derive a conclusion by itself. We expect this to occur due to either accumulated error during the thinking process, or insufficient information inside the model attributed to excess pruning or low-density in training

information. Finally, we aim to quantify the trade-off between thinking tokens and quantisation depth, and measure the delta surplus in allocating higher reasoning budgets in resource-constrained models.

II. Methods

A. Datasets

For quantisation GSM8K [12] is used as a calibration set, which is composed of mostly simple math questions, aiming to preserve accuracy on the complex reasoning questions covered by the MATH benchmark [13], [14] without risking data-leakage. As a second dataset, the discrete reasoning over paragraphs (DROP) dataset is picked, which also requires continuous thinking in order to derive to a reasonable result.

Both benchmark sets are chosen to reflect suitable use cases, and are evaluated on a subset of N=100 samples with cross-validation.

B. Performance metrics

We track full request time, time-per-token (TPT), power draw, inference time, accuracy and F1 as a metric of binary correctness, and recall-oriented understudy for gisting evaluation (ROUGE) and metric for evaluation of translation with explicit ordering (METEOR), which identify incorrect predictions that are too indeterministic to be covered by a regular expression. The metrics are calculated according to:

a) TPT: $\frac{(t_{arrival_t_n} - t_{request})}{\#token}$

b) RougeL: longest common subsequence scoring [15] without stopword removal from the original paper.

c) METEOR: based on unigram-matches it calculates the harmonic mean of unigram-precision and -recall, rewarding well-orderedness of matched words in comparison to the reference passage [16].

d) Power draw: in Watt from `nvidia-smi` real-time system management interface [17].

e) Memory bandwidth: in percent as memory utilization sampled from `nvidia-smi`.

C. Models

As language models we use, at the time of the paper, widely popularised instruct model Qwen3 at 0.6B parameters. Due to missing support of mixture-of-experts (MoE) on Volta-compatible VLLM releases, we decided to evaluate DeepseekR1 1.5B instruct as a second model, despite the potential language or architecture bias this may introduce.

D. Pre- and postprocessing

The MATH dataset uses regular expressions and symbolic verification for substitution and removal of ambiguous but mathematically equivalent latex notation, including removal of units and more. We adopt this approach for this dataset, but add additional removal terms to MATH and DROP due to misclassified responses in the original model due to unnecessary context that leaked through due to their verbose nature.

Besides, we did not have much success using standard prompts for constraining the model to single-valued answers, introducing the need for `<result>value<\result>` prompting and extraction. The evaluation therefore introduces a sequential $O(n)$ overhead per prediction. Due to missing vectorization, this adds a CPU-bound bottleneck that scales linearly with token volume, though remains subordinate to the primary inference latency

E. Inference

We work with VLLM version 0.7.3 for CUDA 12.1 support on NVIDIA Volta V100 GPUs, and substitute for missing thinking token functionality by porting the 0.12.0 VLLM version's functionality. Namely ThinkingBudgetLogitsProcessor for injecting latent `</think>` tokens when exceeding the set thinking budget in tokens. We use this to iterate through configurations of 2^6 to 2^9 tokens, and inject the latent closing thinking token once the limit is reached.

F. Quantisation

According to VLLM documentation, only a fraction of accuracy-preserving quantisation methods are supported [18] on Volta, out of which we chose GPTQ with GPTQModel [10] for managing error rates and for compatibility with our GPUs, instead of the commonly preferred AWQ method [19].

For configurations, we then vary between 4 and 8 bit quantised models.

G. Hardware

We use 1 $\times$ Tesla V100 NVIDIA GPU with 16 GB VRAM for CUDA optimized quantisation and inference, with an attached 60 GB volume of effective 56 GB for code, data, models and dependencies. On a cloud compute instance with Debian 13 Trixie manual installation of extensive dependencies such as CUDA drivers in exchange for a low initial OS size are requred, and comes out at roughly 9.9 GB total usage on the primary, and roughly 40 GB for models and datasets on the second volume.

Running through all 32 configurations of model, dataset, quantisations and thinking budgets on a dataset subset of 100 questions each, we arrive at 2 h and 33 min total, which results in half, 1 h and 16 min, per dataset and configuration. Due to dependency conflicts, separate environments have to be enforced:

a) Quantisation: `gptqmodel==1.4.0`, `triton==2.1.0` and `torch==2.4.0`, `torchvision==0.19.0` with `pytorch-cu121`.

b) Inference: `vllm==0.7.3`, `xformers==0.0.28.post1`, `triton==2.1.0` and `torch==2.5.1`, `torchvision==0.20.1` with `pytorch-cu121`.

H. Overview over tracked information

We run through $\{4, 8 \text{ bit}\} \times \{64, 128, 256, 512 \text{ token}\} \times \{D_1, D_2 \text{ dataset}\} \times \{p_1, p_2 \text{ prompt}\}$, resulting in 64 configurations in total. Per run we then record {TTFT, TPT, Power Draw, Accuracy, F1, ROUGE, METEOR, Memory Bandwidth, Request Time}. These are evaluated in the following section.

III. Evaluation

In the following section we evaluate and interpret our results in terms of model correctness, inference time, token count, memory bandwidth and power draw. The results will be accessible via our repository for further research [20].

A. Correctness

Between allocated token budgets and quantisation levels (Fig. 14) we observe a linearly correlated upward trend with the highest and most reliable scores being at the maximum quantisation depths and token counts.

Compared to the other metrics, METEOR consistently reports lower scores, but also increases slowly from a generally lower baseline.

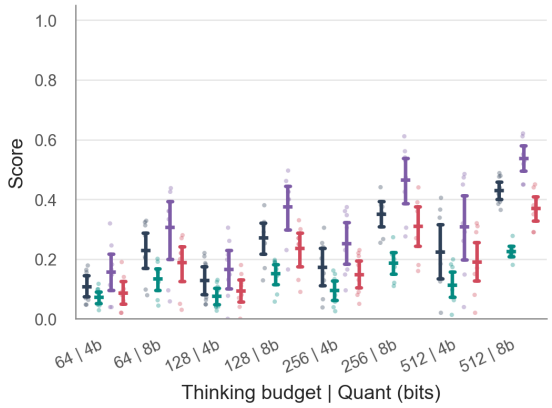

Fig. 3. Correctness metrics acrross multiple thinking budgets (64-512) and quantisation bits (4, 8) per ROUGE (blue), METEOR (green), f1 (purple), accuracy (pink) scores with 95% CI error bars. A clear upward trend for 4 and 8 bit quantisation is visible, with less variability and slightly better performance in higher quantisation settings.

The expansion of the 95% CI error bars at 4 bit quantisation level when scoring higher than 8 bit at half the token budget, reflects its greater stochasticity, undermining model stability.

When comparing the metrics between 4 and 8 bit across all thinking ranges (Fig. 4), the increases encompasses $\leq 10\%$ per step, increasing in similar percentages across all scores.

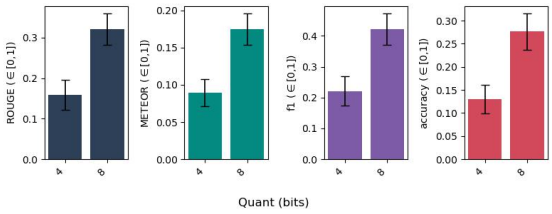

Fig. 4. Aggregated ROUGE, METEOR, accuracy and F1 at 4 and 8 bit quantisation levels with 95% CI error bars each.

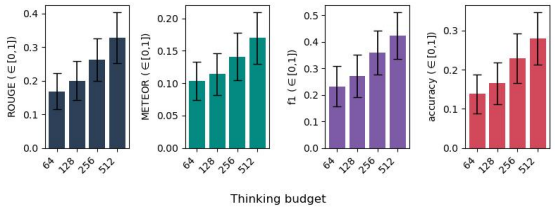

Fig. 5. ROUGE, METEOR, f1 and accuracy across thinking budgets with 95% CI error bars.

The regular increase similarly can be found in various thinking token counts, ranging $\leq 20\%$ for thinking token related correctness gains, in particular for accuracy, as shown in Fig. 4.

According to our recorded performance, the correctness can be linearly estimated (Fig. 5), so in terms of accuracy $0.0365\,q + 0.0003\,\tau - 0.0915$ with $R^2 = 0.571$.

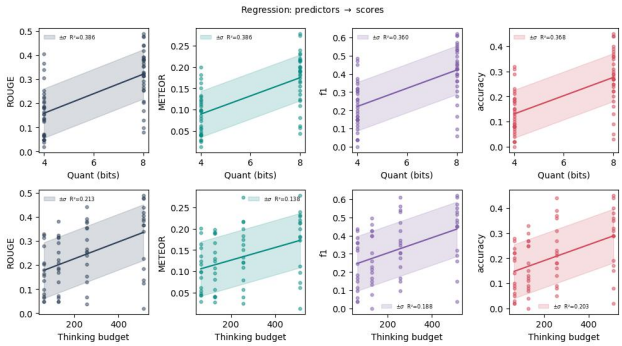

Fig. 6. Linear relationship of score metrics across quantisation levels and thinking budgets with standard deviation and R^2.

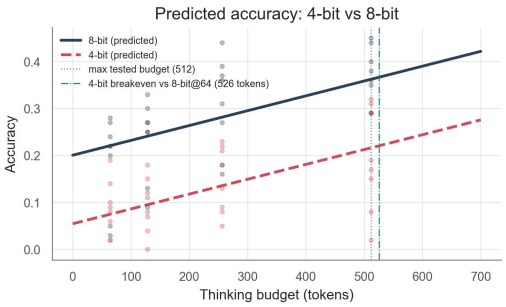

Fig. 7. Crossover accuracy between token count and quantisation level assuming identified linear relationships.

To balance out the accuracy losses in quantised models going from 8 to 4 bit via reasoning, 15% in accuracy need to recovered, leading to additional 462 tokens (Fig. 7).

When trying to generalize compute metrics to linear relationships, we observed a much more nuanced non-gaussian distribution of data, further specified in the following section.

B. Compute

In power, memory bandwidth and time per token, the quantisation level leads to almost unnoticable deviations in memory bandwidth and power draw, despite doubled bit depth. This indicates that the hereby caused overhead is a tiny fraction of the overall power draw and request inference time 8, unlike reasoning token counts.

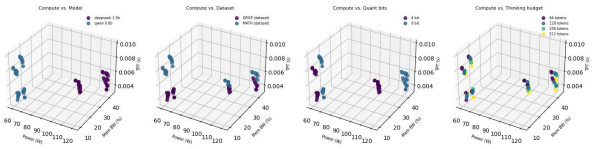

Fig. 8. Decomposition of power draw (W), memory bandwidth (%), time per token (s), model site (parameters), dataset, quantisation levels (bit) and thinking budgets (tokens) ($\mathbf{R} \in [0, 1]$) in 3D space.

Larger model sizes cause the largest consistent jump in power draw and memory bandwidth, while dataset choice only slightly influences time-per-token either due to different triggered reasoning requirements or evaluation script overheads. In the latter, lower token counts contribute to higher TPT (Fig. 8) due to relative higher weight of management overhead per prediction.

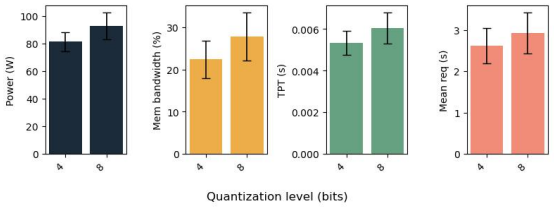

Fig. 9. Compute metrics across quantisation levels with 95% CI error bars.

If in the DROP [21] dataset, the strongly reduced per token inference time is due to easier questions and faster responses, it would be in line with the recorded higher accuracy on this dataset (Fig. 8), despite having used a mathematics-themed calibration set for quantisation.

Similarly, higher quantisation depths lead to higher memory bandwidth and power draw. This leads to believe that the chosen GPU has optimised 8 bit memory accesses, allowing for higher memory bandwidth. Besides, higher accuracy accompanies increased reasoning and powers draw due to correlation in bit depth.

Power and bandwidth across thinking budgets remain roughly the same (Fig. 9), with time per token (TPT) inversely following request time, indicating grouping effects by large token processing at once, or leading request and post-processing overhead. It is possible that a portion of the overhead finds its origin in the additionally implemented, non GPU-optimized post-processing steps in the benchmark evaluation.

The memory bandwidth increases across bits (Fig. 9), appear unintuitive at first, but could be explained through additional management overhead on the GPU side on non 8 bit multiple transfer sizes.

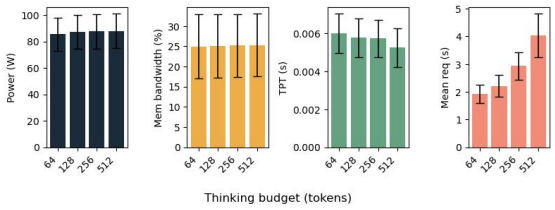

Fig. 10. Compute metrics across thinking budgets with 95% CI error bars.

Conclusively, we summarise, that primarily request time and power draw (as shown in Fig. 10) are most reliably affected of our compute side metrics in varying configuration settings, the first following quantisation levels in a linear relationship, while the latter only appears to be introduced at higher bit depths.

C. Overview

For the final overview we observe the correlations between metrics, and quantify the costs for correctness given our experimentally modulated parameters.

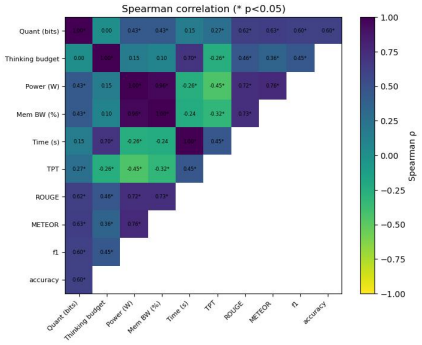

Fig. 11. Heatmap of Spearman correlation of recorded metrics, higher correlation is exhibited from scores to quantisation, than to reasoning tokens.

Looking at the heatmap in Fig. 11, quantisation bits have a higher Spearman correlation with score performance (0.6) than thinking budgets (0.4), which can be explained through the increasing error bar widths in Fig. 14.

It is visible, that the correctness increase from higher bit depths is less power intensive (Fig. 12), but becomes more time-effective as thinking budgets (Fig. 13)increase.

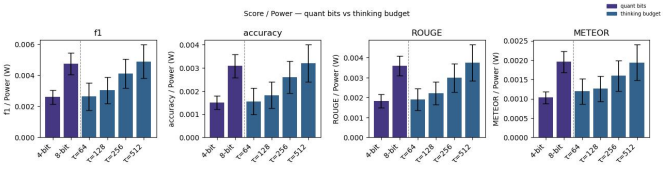

Fig. 12. Score per power draw (W) across quantisation depths (bit) and thinking budgets (tokens) with 95% CI error bars.

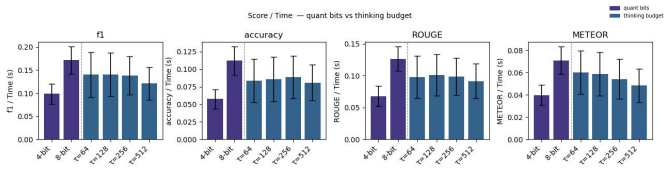

Fig. 13. Score per time (s) across quantisation depths (bit) and thinking budgets (tokens) with 95% CI error bars.

Clearly, quantisation bits are naturally more time effective, whereas reasoning tokens exhibit slightly higher correctness gains with lowered power requirements. This is expected, and in line with our previous findings.

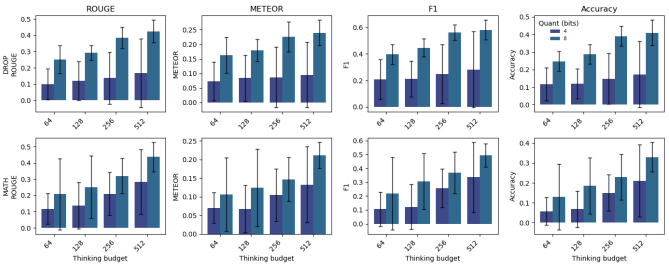

Fig. 14. Comparison of scores between datasets (DROP, MATH) with varying parameter configurations. Highest reliability, in particular for lower thinking budgets, are found in 8 bit quantisation. Higher 4 bit quantisation may reach higher scores but exhibits higher variability in either dataset.

The computed marginal accuracy per quant bit finally lies at $3.6 \cdot 10^{-2}$, and per token at $3 \cdot 10^{-4}$, meaning tokens equivalent to 1 quant bit would come out at 115 tokens. Respectively, marginal power draw per quant bit lies at 2.74 $\frac{W}{bit}$, and per token at $3 \cdot 10^{-3} \frac{W}{token}$. Therefore **power cost per accuracy unit** via quant bits lies at $76.84W/\Delta acc$, higher than via tokens at $9.95 \frac{W}{\Delta acc}$. Meanwhile, our **time cost per accuracy unit** via quant bits lies at $2.38 \frac{s}{\Delta acc}$, much lower than via tokens at $15.46 \frac{s}{\Delta acc}$.

To summarise, our primary takeaway is that increased reasoning token budgets are more power and VRAM efficient, and do in fact increase overall scores, but face overwhelming time tradeoffs. In addition, lower quantisation bits lead to higher variability in correctness, exhibiting worse model reliability.

IV. Discussion

We have shown that higher quantisation can indeed be balanced by increased thinking token counts, while overhead per request overweighs the relative overhead per token at low counts, accordingly lowering relative to larger token budgets. According to our calculations, matching accuracy via quantisation bits costs 7.72× more power than via thinking tokens, while matching via quant bits would cost 0.15× less time than via thinking tokens.

The power draw and memory bandwidths for high and low token requests remain approximately constant, while only quantisation levels raise all recorded compute side performance metrics slightly.

Based on those metrics it is inferred, that not only resource constrained edge devices can make use of increased reasoning tokens with higher quantisation levels, but also systems that can afford longer end-2-end inference times per request. The relationship between token per quantisation bit in accuracy lies at approximately 3.6 hundred token per quantisation bit.

V. Related work

Existing literature already determines a bifurcation in scaling strategies. While standard LLMs benefit from lower-bit configurations for maximising knowledge density, CoT faces a quantisation ceiling due to relying on the stability of long-chain hidden states.

The divergence of quantisation noise suggests that high-fidelity logic cannot yet be effectively compressed in ultra-low bandwidths around 2 quantisation bits without catastrophic failures in complex inference. This indicating that our experiments have not reached the ceiling to significantly demonstrate the edge conditions for significant failures, indicated in our initial hypothesis, leading to expect performance ceilings at only 2 bit and lower quantisation depths.

VI. Limitations

Since our model relies solely on ≤ 1.5B parameter language models, and our token injection sometimes leads to bleeding thinking contents inside the answer sections, it may be possible that different approaches to thinking prompt injection and higher model sizes may have led to slightly better results. It may also be possible that a vectorised evaluation script could have skewed the inference time calculations significantly. Finally, it would have been interesting to observe model behaviour above 512 reasoning tokens, and quantify whether the gains will follow power scaling rules similar to model size scaling laws, or reach a plateau when learnt latent weight-information hits a ceiling. Due to restrictions faced on the admitted compute instances, a further cross-validation and benchmarking of higher token budgets were unable to be started, and are sadly left open to further research.

VII. Conclusion

We conclude that correctness losses in edge-devices with VRAM or compute constraints may be balanced by increased reasoning budgets at constant 4 or 8 quantisation bits and model size at a significant time tradeoff.

Equivalently, accuracy per token is more power-efficient than increased quantisation width, and favors grouping of complex task requests instead of distributing over separate CoT requests due to per-request overhead. Nonetheless, the most reliable and correct configuration set given our experimental setup and benchmarks still lies at 512 token in 8 bit models, favoring both high bit depth and token budgets for high-stake reasoning cases.

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